

Gerald E Butler

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Objective

Seeking a position in any of:

- **Software Engineering and Architecture** using any or all of the following languages: Rust, C/C++, C#, Java, Progress 4GL, Delphi/Pascal, Javascript/Typescript, Go, Dart, SQL, pSql, pgSql, msSql, etc.
- **Enterprise Database modeling, design, and implementation** (both OLTP/3NF+ & OLAP/MDX/Dimensional Modeling) and/or design, development, and implementation of data ETL/ELT workflows with an emphasis on the usage of technologies such as MS-SQL Server, MS-SSIS ETL, Visio, Erwin, etc.

Relevant Technologies

Languages:

Rust, C#, Java, C/C++, Progress 4GL, Delphi/Pascal, many others

Databases:

MS-SQL Server (including MDX), DB2, Oracle, PostgreSQL, SQLite,
MariaDB/MySQL, Mondrian OLAP/MDX

ETL/ELT Tools:

MS-SSIS, Pentaho ETL, Oracle ODI, Informatica

Modeling Tools:

MS-Visio, TOAD, Erwin, MS-SSMS, Eclipse, Spreadsheets/Custom Code to
implementation, Direct DDL Script Authoring

Work History & Experience

Payward Inc./Kraken.com

Software Engineering and Architecture

May 2022 – present

Responsible for porting legacy system in PHP to modern, efficient, memory-safe language (Rust) for financial services (crypto-currency trading platform). System required stringent security, reliability, and latency requirements. After completing porting of software then was responsible for software changes and enhancements related to KYC (Know Your Customer) and AML (Anti-Money Laundering) regulations and requirements. Technologies included, Rust, PHP, MySQL, Redis, PostgreSQL, Kubernetes. Primary focus was Rust development and engineering.

Medical Sabbatical

January 2020 – April 2022

Due to health issues and needing to care for parent I was on a sabbatical for this time period.

Ridge Tool Company/Emerson Incorporated, Elyria, Ohio

Software Architecture, Leadership, and Development

December 2018 – 2020

Lead team consisting of 3 full-time (including myself) developers and up to 4 co-op developers. Java/Spring/Vue.js. MS-SQL Server, DB2, Oracle databases and SSIS and Pentaho Kettle ETL. Agile Scrum/Kanban development processes managed via Rally. Primary responsibilities and accomplishment over the past year consisted of modernizing the development stack *and making the applications more modular by factoring into Java-based "Micro-Services"* with an emphasis on more rapid deployment moving towards a CI process while changing the applications to incorporate newly merged business units. This involved extensive re-modeling of the databases to both support legacy code (through legacy views etc) and new development using proper 3NF OLTP modeling of the databases. Updating security infrastructure to modernize around use of *OpenID, OAuth, and custom authentication and authorization methods* across suite of Web Services and Micro-Services. Extensive mentoring and training of Co-Ops to make them productive, contributing members of the development team.

Oxford & Associates / Laird Technologies, Warren, Ohio Software Development & Architecture Consulting/Contracting July 2018 – December 2018

Design and develop a cross-platform application using Xamarin/C# for UWP, Android, and iOS that connect to safety devices via Bluetooth to perform configuration and allocation. System uses all best practices for cross-platform Xamarin development. Connects to back-end servers via Rest/RPC. Maintains local database in SQL-Lite using a light-weight ORM framework. Designed for scalability of the UI to run on UWP with laptops, desktops, tablets, phablets, as well as the ability to compile and deploy to Android and iOS devices using the Xamarin framework.

Butler Software LLC., Copley, Ohio Software Development & Architecture Consulting/Contracting March 2017 – April 2019

Perform software design and development of various software systems in Java, C#, and related technologies, including: MS-SQL, DB2, OracleDB, etc. ETL in open source Pentaho/Kettle. Design and develop physical property computation modules and supporting infrastructure in Java for polymer colorant concentrates. Designed, developed, tested, documented, and deployed software in C# running as a service on MS Windows OS to identify, track, and report operational status of radiological Continuous Air Monitoring Systems (CAMS) and associated intermediary hardware for Nuclear Armament Maintenance and Inspection facilities. All work conducted as part of my own software development and contracting company whose clients primarily included for this time period: Thermo-Fisher Scientific and Americhem Inc.

The Software Guild (The Learning House Inc), Akron, Ohio On-Line Software Development Instructor March 2016 – March 2017

Performed instruction and tutoring of both in-person and on-line students in both C# and Java development. Emphasis on proper design patterns, best-practices, and agile methodology including, .Net/MVC, Spring/MVC, DAO/Repository Patterns, MVC Pattern, proper coding conventions, etc. Worked to adapt the in-person curriculum for on-line consumption. Designed and Created supplementary lessons and materials, both written and video tutorials, for consumption of the on-line students.

Americhem Inc, Cuyahoga Falls, Ohio

Software Development Consultant/Contractor

Provato Group

June 2016 – August 2016

Designed and developed batch processing ETL jobs and transformations for managing physical property characteristics of polymer composite formulations for the previously developed software application (see previous engagement with Americhem circa 2014/2015). System designed to export physical property characteristics to multi-paged Excel spreadsheets for update and management and then import then in a controlled manner with full error handling and feedback to update the physical property characteristic parameters of polymer formulations.

PNC Bank, Strongsville, Ohio

Software Development Consultant/Contractor

TEK Systems

September 2015 – March 2016

Designed and built Informatica jobs and transformations for the Federal Operations Reporting project. Designed to take input data from Sharepoint forms and from ServiceNow IT Service Management System. Designed and Created ServiceNow jobs and workflows backed by PowerShell scripts to export ServiceNow data to Informatica processing platform. Designed and created Informatica Jobs, Transformations, and supporting scripts to import data from ServiceNow into the Data Warehouse. Designed and documented database schemas using Erwin data modeling tools.

Ridge Tools Inc (a Division of Emerson Electric Inc), Elyria, Ohio

Software Development Consultant/Contractor

Onshore Momentum/Butler Software LLC

December 2014 – September 2015

Migrated data-warehouse in SQL Server/Pentaho Mondrian to Oracle/OBIEE. Redesigned Dimension and Fact table loading procedures, indexing, layout for optimal performance on Oracle/OBIEE and updated ETL Process (Penaho Kettle) to populate the new structures using efficient mechanisms. Changed many ETL processes to more of an ELT type of process to better accommodate the use to Oracle merge and bitmap/bitmap-join indexes for optimal "Star/Snowflake" join query performance. Designed and implemented security hierarchies and mechanisms for the data-warehouse in OBIEE RPD, etc.

Americhem Inc, Cuyahoga Falls, Ohio

Software Development Consultant/Contractor

Provato Group

November 2014 – May 2015

Designed and developed enhancements to existing application that was built for Americhem by other contractors a few years ago. Application was originally developed using JEE/JbossSEAM technologies and was put together in an extremely unmaintainable fashion. I revamped the application, splitting it into modular projects (JPA/ORM, EAR, Web, etc) and

updated the build to properly build in Eclipse, with ability to run unit tests, debug properly, and be friendlier to maintain for Americhem's internal development staff. Enhanced the application with new functionality, including the ability to interface with 3rd party software via Web Services and the addition of "Physical Property" computation modules with configurable, table-driven logic to compute various physical properties for plastic color concentrate "recipes". Trained Americhem internal staff on proper use of Eclipse, Java JEE/SEAM Development, use of SVN version control, etc. Development conducted using an Agile approach using on-line (Google Docs) shared spreadsheets to track user stories, tasks, progress, etc.

Cleveland Clinic Epilepsy Center, Cleveland, Ohio

Software Development Consultant/Contractor

Oxford & Associates

September 2014 – January 2015

Evaluated and documented an existing application written in Tcl/TK using PostgreSQL database back-end to make recommendations as to the ongoing maintenance of the application or redevelopment of the application using internal staff or contractors/vendors. The application had been organically developed internally over 20 years and was the primary software used by the CCF Epilepsy Clinic to manage clinical information with interfaces to the Cleveland Clinic EMR, Billing, and other systems. Recommendations/evaluation took the form of providing detailed cost/benefit analysis and rough estimates of redeveloping the application using modern technologies (Java, .Net, etc) versus continuing to develop the application as a Tcl/TK application.

Ernst & Young, Cleveland, Ohio

Software Development Consultant/Contractor

Onshore Momentum/Butler Software LLC

April 2014 – September 2014

Leading Agile/Scrum development on an application developed in Microsoft Access 2013 and VBA for managing the audit evaluation and review process for the Internal Oversight Audit Team of Ernst & Young. Application designed to import unstructured Audit and Audit Review information from Spreadsheets and an Audit Control System (QMT) for analysis by the Internal Audit Review team. Using Open Source, Pentaho Kettle ETL application for data import and conversion of unstructured data to fully-structured 3NF OLTP database in Access. Migrated the application to use MS-SQL Server and SSIS as well as C#.Net once requirements changed to needing larger number of concurrent users.

Thermo-Fisher Scientific, Oakwood Village, Ohio

Consulting Developer/Technical Lead

Butler Software LLC/AS&E

August 2012 – February 2013

Systems Development Consultant – Radiological Measurement & Scientific Instruments Division, Thermo-Fisher Scientific. Designed and developed 3-D rendering interface using Java OpenGL API's for state-of-health and isotope identification and characterization for the ASP (Advanced Spectroscopic Portal) system.

Ridge Tools Inc (a Division of Emerson Electric Inc), Elyria, Ohio

Software Development Consultant/Contractor

Onshore Momentum/Butler Software LLC

October 2011 – April 2014

Designed and implemented Global Business Intelligence System built around Open-Source technologies and frameworks that use Kettle ETL tools to import BI data from ERP System (JDE/AS400/DB2) and other source systems aggregating data into a “Snowflake/3NF” data warehouse design with optimization aggregations to “Star Schema/Dimensional Modeling”. System designed to keep BI data up-to-date with 15 minute intervals (near real-time analysis capability). Front-end reporting using BIRT reporting technologies, Pentaho Dashboard and Analysis technologies, as well as additional Open-Source add-ons. Developed Operational Dashboard using JQuery and Xmla4JS AJAX libraries performing MDX analysis and presentation against Mondrian OLAP cubes. Oversaw beginnings of conversion of the data warehouse system to Oracle OBIEE due to the parent corporation's decision to move to Oracle technologies. Managed entire process using Agile/Scrum supported by Rally SAAS Scrum software planning tool. Mentored and taught inexperienced developers and co-ops on the necessary technologies, techniques, theory, and practice of data warehouse design and development.

CSI Infusion and Home Health Services Network, Brecksville, Ohio

IT Manager and Lead Software Architect

January 2009 – October 2011

Managed IT Department and software development for Infusion Pharmacy and Home Health Care network business consisting of approximately 150 employees. Oversaw the support of all computer and telecommunications systems including a 3 member IT support staff. Oversaw and participated in all software development of software for managing the Hospital Out-Take to Home Health Care In-Take with a team of 5 software developers. Software designed to provide work-flow for hospital discharge faxed documents, phone calls, e-mails and other discharge related communications through to all communications with the home health agencies, the in-house infusion pharmacy, the patients, and the patients' insurance carriers. Development of software in Java/JEE, SEAM, MS SQL Server, and other frameworks and databases including Pentaho Kettle ETL, SSIS, DTS, etc.

National City Bank: Technical Lead/Developer

4TechWork

August 2008 – January 2009

Systems Development Consultant – Customer Channel Architecture Team, National City Bank – Strongsville, Ohio

Designed, Developed, and Implemented JBoss/JEE-based transaction processing system, running on Redhat Enterprise Linux, with a C#/.Net client on MS-Windows workstations, for guaranteed delivery of document images and associated transaction information in the X9.37/Check21 format for 15,000 workstations nationwide with a peak transaction volume of 4 million transactions per day at a rate of approximately 150 transactions per second. Oversaw the development efforts of a team of 3 developers (including self) and produced, presented, and defended technical design documentation for said system.

Thermo-Fisher Scientific: Consulting Developer/Technical Lead

Technical Search Consultants (Onshore Momentum)/AS&E

June 2007 – January 2009

Systems Development Consultant – Radiological Measurement & Scientific Instruments Division, Thermo-Fisher Scientific

Designed, developed, provided technical and development leadership and supported Java based “Supervisory Control Station” for port traffic scanning radiological detection system for deployment by DHS (Department of Homeland Security)/DNDO (Domestic Nuclear Detection Office)/CBP (Customs and Border Patrol). MOA (Message Oriented Architecture) based around extensive use of JAXB/XML binding and ActiveMQ™ JMS. Java SWING based user interface with modular, pluggable design.

**Sterling Jewelers, Inc: Project Manager, Project Leader, Senior Developer
January 2000 – February 2007**

Project Manager, Store Systems Group – 2004-2007; Project Leader, Store Systems Group – 2002-2004; Senior P/A, Store Systems Group – 2001-2002; Programmer/Analyst, Store Systems Group – 2000-2001

- Responsible for creating, maintaining, and supporting POS software created in Progress® 4GL/RDBMS, C, Perl, Python and Shell-Script for retail store chain of 1,300+ stores (3000+ terminal) nationwide
- Created print driver subsystem for updated POS terminal slip printers written in "C" and Progress 4GL
- Specified, Designed, Developed, and Implemented piece tracking (high-visibility micro-inventory) system for stores systems which resulted in approximately \$10 million dollar reduction in inventory shrink
- Converted store systems and software from SCO OpenServer® to RedHat/Fedora/Ubuntu GNU/Linux™ systems
- Created, developed, maintained, and supported document imaging system for a workload of approximately 50,000 document images per day using Kofax® OCR software, VB.Net, MS-SQL Server using extensive T-SQL stored procedures and other ancillary technologies

Pervasive use (architecture, design, development, support, and maintenance) of all the following disparate technologies:

- Shell Scripting (bash, ksh, csh, zsh, sh, etc)
 - Software Installations and Upgrades in a Unix/Linux environment
 - Diagnostic Data Collection and Analysis
 - Inter-System communications and message delivery
 - Inter-Process coupling
- XML, XPath, XSLT, XQuery
 - Created AJAX library and Web-Client development framework
 - Complex Data-Pattern Business rules in Xpath/XQuery
 - Data Format conversions
 - Inter-Process/Inter-System messaging
- JavaScript/ECMA Script
 - Created AJAX library and toolkit
 - 3-Tier Intra-Net applications with cross-browser support

- Business-Rule engine using "Rhino" JavaScript compiler/interpreter for Java in a Java/J2EE environment
- Perl/Python
 - Automated Telnet, FTP, SSH
 - Created automated "Regression Test" suite for verifying new release of Store Systems software written in Progress 4GL and other ancillary languages
 - Diagnostic and System Health data-collection and reporting
 - Code analysis, reporting, restructuring, and documentation
 - Rapid Prototyping for communications infrastructure
 - Modem (and other communications channels) negotiations and process control arbitration
 - Telnet/Device Interface multiplexer
 - Release generation and management
 - Postscript Printer API and Interface for Report generation
- LISP, Scheme, REP
 - GUI/Kiosk lock-down and control
 - Code formatting and editing
 - Control Automation
 - Business Rule Engine (GUILF)
 - Web-Cam integration with Repair Shop systems for Repair Merchandise tracking and controls
- Progress 4GL
 - Retail POS system
 - Print Device Driver and API
 - Communications Arbitration
- SQL, Oracle, MS-SQL Server
 - Document capture, OCR, and Archive system involving extensive use of Oracle PL/SQL and MS-SQL Server/T-SQL queries, stored procedures, triggers, stored functions, and command-line processes
 - General Purpose "Mid-Tier" for Java/J2EE environment to facilitate rapid development of "Thin-Client", "Browser-Based", "AJAX enabled" intra-net business applications
 - Shipping business rule system
 - B2B communications systems using PL/SQL processes
 - T-SQL stored procedures and functions for cross-database (MS-SQL Server -> Oracle) communications and reconciliation
- Java, J2EE
 - Shipping HUB system for 1,300 nationwide store systems for shipping selection and optimization via FedEx, UPS, DHL/Airborne, courier, armored carrier, U.S.

- Postal Service, or other carriers
 - General Purpose “Mid-Tier” for Java/J2EE environment
 - ”Gift Card” fulfillment system interfacing stores to 3rd Party provider and in-house corporate databases
 - JavaScript accessible Java applets for UDP communication between browser and corporate mainframe COBOL applications
 - Business rule engine using “Rhino” JavaScript compiler/interpreter for Java
 - JSP, EJB, Java Beans, Applets, command-line applications, and services (daemons)
- C/C++
 - Communications HUB for UDP, TCP/IP message based traffic between store systems and corporate systems
 - Serial Device Drivers
 - Linux Kernel debugging, troubleshooting, and modification (device drivers)
 - Telnet and Hardware Device multiplexer
 - Web-Cam control software (Open Source “Camorama”)
 - Galeon (Mozilla/Gtk/Gnome based) web-browser
 - TCP/UDP Servers and Daemons
 - Multi-threaded and Multi-process applications
 - Signal-Handling, IPC, and Share Memory
 - Troubleshoot, debug, patch, and contribute to “Open Source” software for customized usage
- Visual Basic, VB-Script, VBA, VB.Net
 - Document OCR/Archive system interfacing to Oracle, MS-SQL Server, and PostgreSQL databases
 - SQL query and data-analysis from MS-Excel
 - MS-Word/Excel automation

Curtis Software, Inc: Project Leader/Senior Developer

August 1996 – January 2000

Programmer/Analyst, Project Leader, Support Technician, Network Technician

- Supported, Maintained, Designed and Coded Enhancements for, and managed team for Medical Billing application developed in Progress® 4GL/RDBMS
- Approximately 500 clients nationwide ranging from single-user to 20 user systems
- Approximately \$7.5 million in revenue
- Also included significant Delphi and SQL development and specification.

Kent State University: Software Developer/Network Technician

1994-1996

Programmer/Analyst and Network Technician - Managed departmental network for Residence Services department. Created, implemented, and maintained self-serve room assignment system developed using CA-Clipper 4GL and "C" for all university residence halls (~30 halls/~3,000 rooms) for an on-campus population of 15,000+.

U-Haul: Assistant Manager

1993-1994

Truck Maintenance, Hitch Installation, Store Management

U.S. Army: Soldier/HAWK ICC Technician

1989-1993

HAWK Information Coordination Central Technician (Electronics/Missile Systems) - Diagnosed, Repaired, Maintained, and Inspected surface-to-air missile systems including associated CWAR and HPIRR radar systems, TDECC tactical control systems, missile launchers and associated equipment. Included troubleshooting and repairing high-voltage, low-voltage, AC/DC, digital and analog, waveguides, synchronous motor controllers, and hydraulic systems. Spent 1 year working as Brigade Material Operations Facilitator where was responsible for coordinating high-priority acquisition of missile systems components from CONUS (Continental United States) for REDLINE missile batteries within the brigade. Created software to track missile systems outages and RED-TIME and continuously query USAREUR (U.S. Army Europe) as well as CONUS logistics systems for component tracking status and ETA.

Awards

2002 - Achievement of Excellence Award, Sterling, Inc., Akron, Ohio

1994 - President's List, Kent State University, Honors College, Kent, Ohio

1989-1993 - Thrice Awarded Army Achievement Medal, United States Army

Interests

Mathematics, Philosophy, Guitar, Music Theory, Hiking, Swimming, Technology News, Free and Open-Source Software, Linux, Ubuntu, software development, Molecular Biology, Physics

Education

1994-1996: Kent State University, Kent, Ohio

Worked towards BS in Computer Science/Mathematics in the Honors College (2+ years completed)

1989-1993: United States Army

Basic Military Training - 8 weeks, HAWK Information Coordination Central Electronics Technician School - 45 weeks, Primary Leadership Development - 6 weeks, Combat Lifesaver - 2 weeks,

PIP III (secret) - 12 weeks

1985-1989: McKinley Sr. High School, Canton, Ohio
High School College Preparatory Diploma

Professional References

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